



Mall-Connected Shopping & Errand Planner

Final Project Presentation — Full-Stack JavaScript & Mobile Development

React Native • Expo SDK 57 • Supabase PostgreSQL • AsyncStorage

The Problem with Traditional Errand Planning

[!] Disconnected Shopping Notes

Generic note apps lack location and store context, leaving shoppers without store-level item organization.

[!] Budget Surprises at Checkout

Without automatic item price tallies, shoppers lose track of estimated costs until reaching the cashier counter.

[!] Inefficient Mall Travel

Walking back and forth between supermarkets and department stores due to unorganized shopping checklists.

[!] Missed Errand Deadlines

Lack of calendar scheduling and native alarm alerts leads to forgotten shopping trips and errand delays.

The Solution: A Mall-Connected Errand Hub

Mall & Store Directory

Links shopping items directly to Pampanga malls (SM City Clark, Marquee Mall, Nepo Mall, Newpoint Mall) and specific retail stores for targeted trips.

Live Cost Estimation in PHP

Calculates automatic running totals formatted in standard Philippine Peso currency (₱0,000.00) so you always know your budget before checkout.

Calendar Trip Planning

Integrated monthly calendar view to schedule shopping trips by date, view upcoming errands, and set exact native device alarms.

Offline-First Resilience

Loads checklists instantly from local device storage using AsyncStorage with background Supabase PostgreSQL synchronization.

Application Walkthrough & User Flow

1. Home Dashboard

Adaptive two-column grid showing all shopping lists with scroll-driven header, live search, and multi-criteria sorting (Date, Name, Last Opened).

2. List & Item Editor

Create checklists with custom colors, category tags, item quantities, units (pcs, kg, packs), store assignments, and automatic total calculations.

3. Errand Calendar

Interactive monthly calendar displaying scheduled errand dates; tapping any day filters the errands and checklists planned for that trip.

4. Notification Alarms

Native device alarms scheduled via Expo Notifications alerting shoppers ahead of time, surviving device reboots.

Full-Stack Architecture & Technology Stack

FRONTEND MOBILE

React Native 0.86 & Expo SDK 57

File-based navigation with Expo Router, atomic design system (atoms, molecules, organisms), Safe Area Context, and custom ThemeContext (Light & Dark modes).

BACKEND & AUTH

Supabase & PostgreSQL

Relational database modeling malls, stores, lists, and items with Row-Level Security (RLS) policies guaranteeing multi-tenant data privacy.

STORAGE ENGINE

AsyncStorage Cache

Stale-While-Revalidate caching pattern delivering instant offline launch times with seamless background PostgreSQL synchronization.

DEVICE APIS & BUILD

Expo Notifications & EAS

Native exact alarm triggers (SCHEDULE_EXACT_ALARM), Expo Image Picker for avatars, and EAS Build generating standalone Android APKs.

AI Pair-Programming & Developer Ownership

How Claude Was Utilized

- Interactive pair-programming assistant for code reviews and architectural advice.
- Refactoring tricky algorithmic state transitions and notification channel setups.
- Generating initial seed templates and EAS build profiles.
- All code was reviewed, debugged, and verified under real-device testing.

What I Wrote Myself (>20%)

- Atomic UI component system (atoms, molecules, organisms) in Navy (#1B2A4A) & Gold (#D4AF37).
- PostgreSQL relational schema and curated Pampanga mall directories (SM Clark, Marquee, Nepo).
- Expo Router file-based navigation, modal sheets, and safe-area responsive layouts.
- Detailed in AI-USAGE.md with exact GitHub commit hashes.

Technical Challenges & AI Bugs Caught

[!] Scroll Jitter in Animated Header (Commit f176920)

Bug: AI attempted to animate layout height (48 -> 0) and vertical margin inside FlatList, forcing Yoga engine re-measure feedback loops.

Fix: Refactored to non-layout opacity (1 -> 0) and scale transforms, keeping layout dimensions fixed for solid 60 FPS scrolling.

[!] EAS Asset Schema Validation in app.json (Commit c4af20e)

Bug: AI configured icon and adaptivelcon pointing to a .jpg file, failing Expo schema validation during cloud Gradle build.

Fix: Audited project assets with expo-doctor and updated app.json to 1024x1024 .png format, restoring 21/21 passing checks.

[!] Auth Deep Linking & Terms Validation (Commits 1337761 & 24845d4)

Bug: AI missed Supabase PKCE query code (?code=...) deep linking exchange, leaving verified users unable to log in; nested Pressable caused double-toggle checkbox glitch.

Fix: Handled PKCE exchangeCodeForSession with auto-redirect to Home, added 1-tap resend alert, and enforced atomic Terms & Policies validation.

Future Roadmap: What is Next for Plan_.ed

In-Mall Floor Navigation

Step-by-step indoor mapping to locate stores inside SM City Clark, Marquee Mall, and Nepo Mall directly on the screen.

Barcode & Receipt Scanning

Camera integration using Expo Camera to scan product barcodes for instant price lookup and automatic checklist checking.

Real-Time Shared Lists

Collaborative shopping lists enabling family members or errand partners to edit lists simultaneously via Supabase Realtime.

Smart Errand Route Optimizer

Automated sorting algorithm that arranges checklist items based on physical store proximity to minimize walking distance.